Q1. JEE Main 2026 (21 January Shift 2)

Keeping the significant figures in view, the sum of the physical quantities 52.01 m, 153.2 m and 0.123 m is :

(1) 205.3 m (2) 205.333 m (3) 205.33 m (4) 205 m

Q2. JEE Main 2026 (28 January Shift 2)

Match List - I with List - II.

	List - I		List - II
$A.$	Coefficient of viscosity	$I.$	$[ML^{-1}T^{-1}]$
$B.$	Surface tension	$II.$	$[ML^{0}T^{-2}]$
$C.$	Pressure	$III.$	$[ML^{-1}T^{-2}]$
$D.$	Surface energy	$IV.$	$[ML^{2}T^{-2}]$

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-I, C-II, D-III

(3) A-I, B-II, C-IV, D-III

(4) A-I, B-III, C-II, D-IV

Q3. JEE Main 2026 (22 January Shift 2)

If ϵ, E and t represent the free space permittivity, electric field and time respectively, then the unit of $\frac{\epsilon E}{t}$ will be :

(1) Am^2 (2) Am (3) $\mathrm{A/m}^2$ (4) A/m

Q4. JEE Main 2026 (22 January Shift 1)

Match the LIST-I with LIST-II

	List-I		**List-II**
$A.$	Spring constant	$I.$	$\mathrm{ML^2T^{-2}K^{-1}}$
$B.$	Thermal conductivity	$II.$	$\mathrm{ML^0T^{-2}}$
$C.$	Boltzmann constant	$III.$	$\mathrm{ML^2T^{-3}A^{-2}}$
$D.$	Inductive reactance	$IV.$	$\mathrm{MLT^{-3}K^{-1}}$

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-I, B-IV, C-II, D-III

(3) A-II, B-I, C-IV, D-III

(4) A-III, B-II, C-IV, D-I

Q5. JEE Main 2026 (23 January Shift 2)

A ball of radius r and density ρ dropped through a viscous liquid of density σ and viscosity η attains its terminal velocity at time t, given by $t = A\rho^a r^b \eta^c \sigma^d$, where A is a constant and a, b, c and d are integers. The value of $\frac{b+c}{a+d}$ is ____.

Q6. JEE Main 2026 (21 January Shift 1)

Consider a modified Bernoulli equation.

$\left(\mathrm{P} + \frac{A}{Bt^2}\right) + \rho g(h + Bt) + \frac{1}{2}\rho V^2 = \text{constant}$

If t has the dimension of time then the dimensions of A and B are ____, ____ respectively.

(1) $\left[\mathrm{ML^0T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-2}}\right]$

(2) $\left[\mathrm{ML^0T^{-2}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

(3) $\left[\mathrm{ML^0T^{-1}}\right]$ and $\left[\mathrm{M^0LT}\right]$

(4) $\left[\mathrm{ML^0T^{-1}}\right]$ and $\left[\mathrm{M^0LT^{-1}}\right]$

Q7. JEE Main 2026 (24 January Shift 1)

Match the LIST-I with LIST-II

	List-I		List-II
A.	Magnetic induction	I.	$\mathrm{M\,L\,T^{-2}\,A^{-2}}$
B.	Magnetic flux	II.	$\mathrm{M\,L^2\,T^{-2}\,A^{-2}}$
C.	Magnetic permeability	III.	$\mathrm{M\,L^0\,T^{-2}\,A^{-1}}$
D.	Self inductance	IV.	$\mathrm{M\,L^2\,T^{-2}\,A^{-1}}$

Choose the correct answer from the options given below:

(1) A-III, B-IV, C-II, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-I, B-III, C-IV, D-II

ANSWERS AND SOLUTIONS

1. (1) **2.** (1) **3.** (3) **4.** (1) **5.** 1 **6.** (4) **7.** (2)

1. **(1)** When adding measurements, the result must be limited to the least number of decimal places among all quantities.
The measurements are: 52.01 m (2 decimal places), 153.2 m (1 decimal place), and 0.123 m (3 decimal places).
The limiting factor is 1 decimal place.
Calculating the sum: 52.01 + 153.2 + 0.123 = 205.333 m.
Rounding to 1 decimal place gives 205.3 m, which is option (1).

2. **(1)** (A) Coefficient of viscosity:
$\eta = \frac{F \cdot dr}{A \cdot dv} = \frac{[MLT^{-2}][L]}{[L^2][LT^{-1}]} = [ML^{-1}T^{-1}] \rightarrow$ IV
(B) Surface tension:
$S = \frac{F}{L} = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}] = [ML^0T^{-2}] \rightarrow$ III
(C) Pressure:
$P = \frac{F}{A} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}] \rightarrow$ I
(D) Surface energy:
$E = S \times A = [MT^{-2}][L^2] = [ML^2T^{-2}] \rightarrow$ II
Hence, A-IV, B-III, C-I, D-II.

3. **(3)** We need to find the unit of $\frac{\epsilon E}{t}$.
From Maxwell's equations, the displacement current density is given by $J_d = \epsilon \frac{\partial E}{\partial t}$.
The term $\frac{\epsilon E}{t}$ has the same dimensions as displacement current density J_d.
Current density J is defined as current per unit area, $J = \frac{I}{A}$.
The SI unit of current I is Ampere (A) and the SI unit of area A is m^2.
Therefore, the unit of current density is A/m^2.
Alternatively, using dimensions:
$[\epsilon] = M^{-1}L^{-3}T^4A^2$
$[E] = MLT^{-3}A^{-1}$
$[t] = T$
Dimension of $\frac{\epsilon E}{t} = \frac{(M^{-1}L^{-3}T^4A^2)(MLT^{-3}A^{-1})}{T} = L^{-2}A$.
This corresponds to the unit A/m^2.

4. **(1)** A. Spring constant: $F = kx \Rightarrow [k] = MLT^{-2}/L = ML^0T^{-2} \rightarrow$ II
B. Thermal conductivity: $Q/t = KA(dT/dx) \Rightarrow [K] = ML^2T^{-3}/(L^2 \cdot K/L) = MLT^{-3}K^{-1} \rightarrow$ IV
C. Boltzmann constant: $E = k_BT \Rightarrow [k_B] = ML^2T^{-2}K^{-1} \rightarrow$ I

D. Inductive reactance: same dimension as resistance $= ML^2T^{-3}A^{-2} \rightarrow$ III

Matching: A-II, B-IV, C-I, D-III.

5. **(1)** Using dimensional analysis on $t = A\rho^a r^b \eta^c \sigma^d$:

the dimensions are [T] = $[ML^{-3}]^a[L]^b[ML^{-1}T^{-1}]^c[ML^{-3}]^d$.

Expanding: [T] = $M^{a+c+d}L^{-3a+b-c-3d}T^{-c}$.

Comparing powers: For T: $1 = -c$, giving $c = -1$.

For M: $0 = a + c + d$, so $a + d = 1$.

For L: $0 = -3a + b - c - 3d = -3a + b + 1 - 3d$, giving $b = 3(a + d) - 1 = 3(1) - 1 = 2$.

Therefore $(b + c)/(a + d) = (2 - 1)/1 = 1$.

6. **(4)** All terms must have dimension of pressure $[ML^{-1}T^{-2}]$.

For $(h + Bt)$: Since ρgh has dimension of pressure, $[h] = [L]$.

For dimensional consistency: $[Bt] = [L]$, so $[B] = [LT^{-1}] = [M^0LT^{-1}]$

For $\frac{A}{Bt^2}$: $\frac{[A]}{[B][T^2]} = [ML^{-1}T^{-2}]$

$[A] = [B][T^2][ML^{-1}T^{-2}] = [LT^{-1}][T^2][ML^{-1}T^{-2}] = [ML^0T^{-1}]$

$[A] = [ML^0T^{-1}]$ and $[B] = [M^0LT^{-1}]$

7. **(2)** Match physical quantities with dimensional formulas using fundamental principles.

Magnetic Induction (A): From $F = qvB$, $[B] = \frac{MLT^{-2}}{AT \cdot LT^{-1}} = MT^{-2}A^{-1} =$ III

Magnetic Flux (B): $\Phi = BA$, $[\Phi] = MT^{-2}A^{-1} \cdot L^2 = ML^2T^{-2}A^{-1} =$ IV

Magnetic Permeability (C): From $B = \mu H$ with $[H] = ALT^{-1}$, $[\mu] = \frac{MT^{-2}A^{-1}}{AL^{-1}T^{-1}} = MLT^{-2}A^{-2} =$ I

Self Inductance (D): From $\varepsilon = L\frac{dI}{dt}$, $[L] = \frac{ML^2T^{-3}A^{-1}}{AT^{-1}} = ML^2T^{-2}A^{-2} =$ II

Correct matching: A-III, B-IV, C-I, D-II